

ADV: Functions (Adv), F1 Working with Functions (Adv)
Surds and Rounding (Y11)

Teacher: Ms Najem

Exam Equivalent Time: 60 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Questions

1. Functions, 2ADV F1 2018 HSC 1 MC

What is the value of $7^{-1.3}$ correct to two decimal places?

- (A) 0.07
- (B) 0.08
- (C) -12.54
- (D) -12.55

2. Functions, 2ADV F1 2012 HSC 1 MC

What is **4.09784** correct to three significant figures?

- (A) **4.09**
- (B) **4.10**
- (C) **4.097**
- (D) **4.098**

3. Functions, 2ADV F1 2012 HSC 2 MC

Which of the following is equal to $\frac{1}{2\sqrt{5} - \sqrt{3}}$?

- (A) $\frac{2\sqrt{5} - \sqrt{3}}{7}$
- (B) $\frac{2\sqrt{5} + \sqrt{3}}{7}$
- (C) $\frac{2\sqrt{5} - \sqrt{3}}{17}$
- (D) $\frac{2\sqrt{5} + \sqrt{3}}{17}$

4. Functions, 2ADV F1 2014 HSC 1 MC

What is the value of $\frac{\pi^2}{6}$, correct to 3 significant figures?

- (A) **1.64**
- (B) **1.65**
- (C) **1.644**
- (D) **1.645**

5. Functions, 2ADV F1 2015 HSC 1 MC

What is **0.005 233 59** written in scientific notation, correct to 4 significant figures?

- (A) **5.2336×10^{-2}**
- (B) **5.234×10^{-2}**
- (C) **5.2336×10^{-3}**
- (D) **5.234×10^{-3}**

6. Functions, 2ADV F1 2019 HSC 1 MC

What is the value of π^{10} to two significant figures?

- (A) **9.36×10^4**
- (B) **9.4×10^4**
- (C) **9.36×10^5**
- (D) **9.4×10^5**

7. Functions, 2ADV F1 2005 HSC 1a

Evaluate $\sqrt{\frac{275.4}{5.2 \times 3.9}}$ correct to two significant figures. (2 marks)

8. Functions, 2ADV F1 2007 HSC 1a

Evaluate $\sqrt{\pi^2 + 5}$ correct to two decimal places. (2 marks)

9. Functions, 2ADV F1 2007 HSC 1c

Rationalise the denominator of $\frac{1}{\sqrt{3} - 1}$. (2 marks)

10. Functions, 2ADV F1 2004 HSC 1c

Solve $\frac{x-5}{3} - \frac{x+1}{4} = 5$. (2 marks)

11. Functions, 2ADV F1 2005 HSC 1d

Express $\frac{(2x-3)}{2} - \frac{(x-1)}{5}$ as a single fraction in its simplest form. (2 marks)

12. Functions, 2ADV F1 2010 HSC 1b

Find integers a and b such that $\frac{1}{\sqrt{5}-2} = a + b\sqrt{5}$. (2 marks)

13. Functions, 2ADV F1 2011 HSC 1a

Evaluate $\sqrt[3]{\frac{651}{4\pi}}$ correct to four significant figures. (2 marks)

14. Functions, 2ADV F1 2011 HSC 1b

Simplify $\frac{n^2-25}{n-5}$. (1 mark)

15. Functions, 2ADV F1 2011 HSC 1f

Rationalise the denominator of $\frac{4}{\sqrt{5}-\sqrt{3}}$.

Give your answer in the simplest form. (2 marks)

16. Functions, 2ADV F1 2013 HSC 11a

Evaluate $\ln 3$ correct to three significant figures. (1 mark)

17. Functions, 2ADV F1 2014 HSC 11a

Rationalise the denominator of $\frac{1}{\sqrt{5}-2}$. (2 marks)

18. Functions, 2ADV F1 2015 HSC 11c

Express $\frac{8}{2+\sqrt{7}}$ with a rational denominator. (2 marks)

19. Functions, 2ADV F1 2017 HSC 11a

Rationalise the denominator of $\frac{2}{\sqrt{5}-1}$. (2 marks)

20. Functions, 2ADV F1 2018 HSC 11a

Rationalise the denominator of $\frac{3}{3+\sqrt{2}}$. (2 marks)

21. Functions, 2ADV F1 2008 HSC 1a

Evaluate $2\cos\left(\frac{\pi}{5}\right)$ correct to three significant figures. (2 marks)

22. Functions, 2ADV F1 2008 HSC 1e

Expand and simplify $(\sqrt{3}-1)(2\sqrt{3}+5)$. (2 marks)

23. Functions, 2ADV F1 2004 HSC 1a

The radius of Mars is approximately **3 397 000 m**. Write this number in scientific notation, correct to two significant figures. (2 marks)

24. Functions, 2ADV F1 2004 HSC 1d

Find integers a and b such that $(3-\sqrt{2})^2 = a - b\sqrt{2}$. (2 marks)

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Worked Solutions

1. Functions, 2ADV F1 2018 HSC 1 MC

$$\begin{aligned}7^{-1.3} &= 0.0796\dots \\ &= 0.08 \text{ (2 d.p.)} \\ \Rightarrow B\end{aligned}$$

2. Functions, 2ADV F1 2012 HSC 1 MC

$$\begin{aligned}4.10 \\ \Rightarrow B\end{aligned}$$

3. Functions, 2ADV F1 2012 HSC 2 MC

$$\begin{aligned}\frac{1}{2\sqrt{5} - \sqrt{3}} \times \frac{2\sqrt{5} + \sqrt{3}}{2\sqrt{5} + \sqrt{3}} \\ &= \frac{2\sqrt{5} + \sqrt{3}}{(2\sqrt{5} - \sqrt{3})(2\sqrt{5} + \sqrt{3})} \\ &= \frac{2\sqrt{5} + \sqrt{3}}{(2\sqrt{5})^2 - (\sqrt{3})^2} \\ &= \frac{2\sqrt{5} + \sqrt{3}}{17} \\ \Rightarrow D\end{aligned}$$

4. Functions, 2ADV F1 2014 HSC 1 MC

$$\begin{aligned}\frac{\pi^2}{6} &= 1.6449\dots \\ &= 1.64 \text{ (3 sig. figures)} \\ \Rightarrow A\end{aligned}$$

5. Functions, 2ADV F1 2015 HSC 1 MC

$$\begin{aligned}0.005\,233\,59 \\ &= 5.234 \times 10^{-3} \\ \Rightarrow D\end{aligned}$$

6. Functions, 2ADV F1 2019 HSC 1 MC

$$\begin{aligned}\pi^{10} &= 93648.04\dots \\ &= 9.4 \times 10^4 \\ \Rightarrow B\end{aligned}$$

7. Functions, 2ADV F1 2005 HSC 1a

$$\begin{aligned}\sqrt{\frac{275.4}{5.2 \times 3.9}} &= 3.685\dots \\ &= 3.7 \text{ (2 sig. figures)}\end{aligned}$$

8. Functions, 2ADV F1 2007 HSC 1a

$$\begin{aligned}\sqrt{\pi^2 + 5} &= 3.8561\dots \\ &= 3.86 \text{ (to 2 d.p.)}\end{aligned}$$

9. Functions, 2ADV F1 2007 HSC 1c

$$\begin{aligned}\frac{1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} \\ &= \frac{\sqrt{3}+1}{(\sqrt{3})^2 - 1^2} \\ &= \frac{\sqrt{3}+1}{2}\end{aligned}$$

10. Functions, 2ADV F1 2004 HSC 1c

$$\begin{aligned}\frac{x-5}{3} - \frac{x+1}{4} &= 5 \\ 12\left(\frac{x-5}{3}\right) - 12\left(\frac{x+1}{4}\right) &= 12 \times 5 \\ 4x - 20 - 3x - 3 &= 60 \\ x - 23 &= 60 \\ \therefore x &= 83\end{aligned}$$

11. Functions, 2ADV F1 2005 HSC 1d

$$\begin{aligned}\frac{(2x-3)}{2} - \frac{(x-1)}{5} \\ &= \frac{5(2x-3) - 2(x-1)}{10} \\ &= \frac{10x - 15 - 2x + 2}{10} \\ &= \frac{8x - 13}{10}\end{aligned}$$

12. Functions, 2ADV F1 2010 HSC 1b

$$\begin{aligned}\text{Given } \frac{1}{\sqrt{5}-2} &= a + b\sqrt{5} \\ \frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} &= \frac{\sqrt{5}+2}{(\sqrt{5})^2 - 2^2} \\ &= \frac{\sqrt{5}+2}{5-4} \\ &= 2 + \sqrt{5} \\ \therefore a &= 2, b = 1\end{aligned}$$

13. Functions, 2ADV F1 2011 HSC 1a

$$\begin{aligned}\sqrt[3]{\frac{651}{4\pi}} &= 3.72783\dots \\ &= 3.728 \text{ (to 4 sig. figures)}\end{aligned}$$

MARKER'S COMMENT: Show answer to 5 or 6 decimals before rounding. Correct rounding of a wrong answer still receives half marks.

14. Functions, 2ADV F1 2011 HSC 1b

$$\begin{aligned}\frac{n^2 - 25}{n - 5} &= \frac{(n-5)(n+5)}{n-5} \\ &= n + 5\end{aligned}$$

15. Functions, 2ADV F1 2011 HSC 1f

$$\begin{aligned}\frac{4}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}} \\&= \frac{4(\sqrt{5}+\sqrt{3})}{5-3} \\&= 2(\sqrt{5}+\sqrt{3})\end{aligned}$$

16. Functions, 2ADV F1 2013 HSC 11a

$$\begin{aligned}\ln 3 &= 1.09861... \\&= 1.10 \text{ (to 3 sig. figures)}\end{aligned}$$

17. Functions, 2ADV F1 2014 HSC 11a

$$\begin{aligned}\frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} \\&= \frac{\sqrt{5}+2}{(\sqrt{5})^2-2^2} \\&= \sqrt{5}+2\end{aligned}$$

18. Functions, 2ADV F1 2015 HSC 11c

$$\begin{aligned}\frac{8}{2+\sqrt{7}} \times \frac{2-\sqrt{7}}{2-\sqrt{7}} \\&= \frac{8(2-\sqrt{7})}{2^2-(\sqrt{7})^2} \\&= \frac{8(2-\sqrt{7})}{4-7} \\&= \frac{-8(2-\sqrt{7})}{3}\end{aligned}$$

19. Functions, 2ADV F1 2017 HSC 11a

$$\begin{aligned}\frac{2}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} &= \frac{2(\sqrt{5}+1)}{(\sqrt{5})^2-1} \\&= \frac{2(\sqrt{5}+1)}{4} \\&= \frac{\sqrt{5}+1}{2}\end{aligned}$$

20. Functions, 2ADV F1 2018 HSC 11a

$$\begin{aligned}\frac{3}{3+\sqrt{2}} \times \frac{3-\sqrt{2}}{3-\sqrt{2}} &= \frac{3(3-\sqrt{2})}{3^2-(\sqrt{2})^2} \\&= \frac{3(3-\sqrt{2})}{7}\end{aligned}$$

21. Functions, 2ADV F1 2008 HSC 1a

$$\begin{aligned}2 \cos\left(\frac{\pi}{5}\right) &= 1.6180... \\&= 1.62 \text{ (3 sig)}\end{aligned}$$

22. Functions, 2ADV F1 2008 HSC 1e

$$\begin{aligned}(\sqrt{3}-1)(2\sqrt{3}+5) \\&= 2 \times 3 + 5\sqrt{3} - 2\sqrt{3} - 5 \\&= 1 + 3\sqrt{3}\end{aligned}$$

23. Functions, 2ADV F1 2004 HSC 1a

$$\begin{aligned}3\,397\,000 \\&= 3.397 \times 10^6 \\&= 3.4 \times 10^6 \text{ m (2 sig figures)}\end{aligned}$$

24. Functions, 2ADV F1 2004 HSC 1d

$$\begin{aligned}(3 - \sqrt{2})^2 &= 9 - 6\sqrt{2} + (\sqrt{2})^2 \\&= 9 - 6\sqrt{2} + 2 \\&= 11 - 6\sqrt{2} \\ \therefore a &= 11, \quad b = 6\end{aligned}$$

